

MAYA ANGIA

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PROFILE

Biomimicry-driven biomedical engineer building AI-enabled systems across medical imaging, robotics, biosensing, and implantable therapeutics. I translate biological principles and clinical problems into computational and physical systems, with experience in deep learning, computer vision, multimodal sensing, embedded control, and end-to-end prototyping.

EDUCATION

Georgia Institute of Technology - B.S. Biomedical Engineering, Minor in AI/ML | GPA: 3.92 | Graduation: Spring 2028
Courses: Biomed ML, Health Wearables/Instrumentation, System Optimization, Computational Biology, Medical Devices

EXPERIENCE

- Solutions Consulting & Engineering Intern** May 2026 – Present
World Wide Technology
• Enterprise security and post-quantum cryptography consulting for Fortune 500 healthcare/medical-device companies
- Drug Delivery Systems Researcher** January 2026 – Present
Abramson Lab, Georgia Tech
• Developing implantable, microscale drug delivery capsules for autonomous opioid overdose detection and reversal
- Biomimicry Robotics Researcher** June 2025 – May 2026
Bhamla Lab, Georgia Tech
• Translated mudskipper locomotion and biological kinematics into an amphibious, water-hopping robot ([APS 2026](#))
• Fabricated and tested DARPA-backed flexible squid-inspired jet nozzles to improve underwater propulsion efficiency
- Neuromorphic Engineering Researcher** June 2025 – August 2025
Raman Lab, Washington University in St. Louis
• Built an automated behavioral and sensor system to study locust olfaction as a model for artificial sensing
• Integrated microcontrollers, motors, olfactometers, and synchronized recording; increased efficiency by over 200%
- AI/ML Cancer Pathology Intern** August 2024 – May 2025
Madabhushi Lab, Emory University
• Built deep learning/computer vision pipeline for ER+ breast cancer prognosis across 10,000+ histopathology WSIs
• Quantified spatial and collagen fiber features as candidate biomarkers of metastatic risk and survival outcomes

PROJECTS

- Foveate – Bio-Inspired Computational Pathology - Independent Research** (github.com/mangia3/foveate)
PyTorch, NVIDIA MONAI, Virchow2, OpenSlide, Hugging Face, CUDA/HPC
• Developing a bio-inspired saccadic attention framework for gigapixel pathology to test whether selective WSI sampling improves diagnostic accuracy per unit compute over uniform sampling.
• Built a reproducible CAMELYON16 pipeline with physical-scale WSI sampling, frozen Virchow2 embeddings, nested compute budgets, provenance-aware caching, leakage safeguards, and deterministic GPU evaluation
- 4sight - 2nd Place Cloudflare Track at TreeHacks 2026** (<https://devpost.com/software/4sight-neoslb>)
• Built a multimodal wearable ML pipeline using HRV, motion, stress, sleep, and visual contextual signals to estimate near-term fatigue, decision-making, and physiological vulnerability and trigger adaptive interventions
• **Designing a v2 study** (github.com/mangia3/4sightV2) of personalized physiological-state forecasting, cross-user adaptation, missing-sensor robustness, and compute-constrained inference
- MedGraph - 1st Place Data Science at HooHacks 2025** (<https://devpost.com/software/medgraph>)
• Built a privacy-preserving federated GNN for bias-aware disease prediction across decentralized hospital EHR data

CONFERENCES/EVENTS: YC Startup School 2026, APS Global Summit 2026 ([presented bio-physics research](#))

SKILLS

Biomedical & Computational: Medical imaging, computer vision, computational biology, biomechanics, biosensing, medical robotics, multimodal sensing
Programming & ML: Python, MATLAB, PyTorch, MONAI, TensorFlow, NumPy, OpenCV, neural networks, federated learning, GPU/HPC
Robotics & Embedded Systems: Raspberry Pi, Arduino, cameras/sensors, embedded control, CAD/Fusion 360, rapid prototyping, hardware-software integration